Member 3 Report: Conditional Probability, Multiplication, Total Probability and Bayes

Task 1, Smart Buildings dataset, Chapter 2. Notebook Sections 5.5 to 5.8 (slides 6 and 7). This report explains every result, every variable and every name used in my part of the notebook `T1_M3_draft.ipynb`.

1. What my part does, in one paragraph

My part asks one connected story about the generated buildings: **do big buildings need extra air handlers (AHUs), and how can we move between "given the building, what happens to AHUs" and "given the AHUs, what is the building"?** Section 5.5 measures the effect of size with conditional probability. Section 5.6 builds joint probabilities with the multiplication and chain rules. Section 5.7 rebuilds the overall probability of extra AHUs from floor-level rates (total probability). Section 5.8 reverses the direction with Bayes' theorem. Every result is checked against a direct count of rows, and all numbers describe the 200 accepted generated buildings, not real buildings.

2. The data behind my part

I use only the analysis table `df` that Member 1 built (200 rows, one row per accepted generated building). I never change it or the raw tables.

Name in the notebook	What it is	Meaning in plain words
`df_raw`	raw buildings table, 200 x 54	untouched original file (not used by me)
`df`	analysis table, 200 rows x 37 columns before the event columns	Member 1's cleaned copy: shorter column names, 17 constant columns removed
`n_zones`	integer column	number of zones (rooms/areas) in the building; min 21, median 64, max 305

Name in the notebook	What it is	Meaning in plain words
`ahu.num_ahus` and `ahus_configured`	integer columns (identical values)	how many air handlers the building was **configured** with (3 to 7)
`min_ahus_required`	integer column	how many AHUs the building **needs**; equals `ceil(n_zones / 10)` in every row
`ahus_added_beyond_config`	integer column	extra AHUs the generator added beyond the configured number
`floor_plan.num_floors`	integer column	number of floors: 2, 3 or 4
`seed`	integer column	generator seed that identifies the building (link key)
`A_large`	Boolean column (Member 1)	True when `n_zones > 64` (strictly greater; 6 buildings with exactly 64 are not large). 98 True
`B_extra_ahu`	Boolean column (Member 1)	True when `ahus_added_beyond_config > 0`. 146 True
`F2`, `F3`, `F4`	Boolean columns (Member 1)	True when the building has 2, 3 or 4 floors: 74, 63 and 63 buildings
`O2`	Boolean column (Member 1)	2 occupants per zone (108 buildings). Not used in my part

The three most important facts: **64 is the median zone count**, **146 of 200 buildings needed extra AHUs**, and the generator's rule is *required AHUs = ceil(zones / 10)*, while only 3 to 7 AHUs are configured.

3. Notation used throughout (events and symbols)

Symbol	Notebook name	Meaning
A	`A` (= `df.A_large`)	the building is large (more than 64 zones)

Symbol	Notebook name	Meaning
not A	`~A`	the building is not large (64 zones or fewer)
B	`Bx` (= `df.B_extra_ahu`)	the building needed at least one extra AHU
not B	`~Bx`	no extra AHU was needed
F2, F3, F4	`df.F2`, `df.F3`, `df.F4`	exactly 2, 3 or 4 floors. General form F_k, k = 2, 3, 4
P(X)	`.mean()` of a True/False column	share of the 200 buildings where X is True
P(X and Y)	`(X & Y).mean()`	share where both are True (the intersection)
P(X given Y), written P(X | Y)	`Pc(X, Y)`	among only the buildings where Y is True, the share where X is True
N(X)	`.sum()`	the count of buildings where X is True

Key idea behind every calculation. The 200 buildings are treated as a finite population of equally likely outcomes, so every probability is a count ratio such as 97/200. "Conditioning on A" simply means looking only at the 98 rows where A is True and recomputing the share inside that smaller group. All results are conditional on the generator having accepted the seed.

4. Every variable created in my cells

Section 5.5 (conditional probability)

Name	What it holds	Value / note
`plt`	matplotlib plotting module	used for all charts
`isclose`	function from `math`	tests that two decimal numbers are equal up to rounding; used in the `assert` checks

Name	What it holds	Value / note
`Pc(a, b)`	function I defined	returns P(a given b) = count(a and b) / count(b) for two True/False columns
`A`, `Bx`	shortcuts to `df.A_large`, `df.B_extra_ahu`	the two events
`n`	`len(df)`	200, the total number of buildings
`counts_AB`	2 x 2 table of counts with totals	see Section 5 below
`P_A`	P(A)	98/200 = 0.4900
`P_B`	P(B)	146/200 = 0.7300
`P_AB`	P(A and B)	97/200 = 0.4850
`P_B_given_A`	P(B given A)	97/98 = 0.9898
`P_B_given_notA`	P(B given not A)	49/102 = 0.4804
`rows`, `threshold_check`	list and table for the threshold test	see Section 5 below
`t`, `big`	loop variable and True/False column inside the threshold loop	`big` is "n_zones > t" for the current threshold `t`. Local only, `df` is not changed
`need_more`	True/False column: `min_ahus_required > ahus_configured`	proves B is exactly "required AHUs exceed configured AHUs"
`big70`	True/False column: `n_zones > 70`	buildings with more than 70 zones
`fig`, `ax`, `labels`, `vals`, `bars`	chart objects	the bar chart

Section 5.6 (multiplication and chain rule)

Name	What it holds	Value
`F4`	shortcut for `df.F4`	63 four-floor buildings
`joint_rule_1`	P(A) x P(B given A)	0.4850
`joint_rule_2`	P(B) x P(A given B) (other order)	0.4850
`joint_direct`	direct count of A and B	97/200 = 0.4850
`joint_if_independent`	P(A) x P(B), what the plain product would give	0.3577
`p1`	P(F4)	63/200 = 0.3150
`p2`	P(A given F4)	50/63 = 0.7937
`p3`	P(B given F4 and A)	50/50 = 1.0000
`chain`	p1 x p2 x p3 (chain rule)	0.2500
`chain_direct`	direct count of F4 and A and B	50/200 = 0.2500
`multiplication_table`	summary table rule vs direct	2 rows

Section 5.7 (total probability)

Name	What it holds	Value
`floor_events`	dictionary {2: F2, 3: F3, 4: F4}	the partition pieces
`pF`	dictionary of P(F_k)	{2: 0.370, 3: 0.315, 4: 0.315}
`pB_F`	dictionary of P(B given F_k)	{2: 0.5270, 3: 0.7778, 4: 0.9206}
`pB_total`	sum of P(B given F_k) x P(F_k)	0.7300
`total_prob_table`	table of the pieces	see Section 7 below
`pB_total_A`	same result from the A / not A partition	0.7300

Name	What it holds	Value
`joint_F_B`	table of P(floors and B) for each floor count	see Section 7 below

Section 5.8 (Bayes) and hand-off

Name	What it holds	Value
`post`	dictionary of P(F_k given B), the posterior	{2: 0.2671, 3: 0.3356, 4: 0.3973}
`post_F4`	P(F4 given B)	0.3973
`direct_post`	the same posteriors by direct count among B buildings	identical to `post`
`pnotB`	P(not B) = 1 - `pB_total`	0.2700
`post_notB`	dictionary of P(F_k given not B)	{2: 0.6481, 3: 0.2593, 4: 0.0926}
`prior_posterior`	table prior vs likelihood vs posterior	see Section 8 below
`post_A_given_B`	P(A given B) by Bayes	0.6644
`x`, `w`	bar positions and bar width for the grouped chart	chart helpers
`arrow`, `floor_y`, `fy`, `ly`, `p_cond`, `lab`, `col`	helpers for the probability tree drawing	chart helpers
`m3_results`	dictionary of all rule-based results	handed to Member 4 for validation

Names such as `fig`, `ax`, `x`, `w`, `rows` are throw-away working names. The names that matter for the team are `Pc`, `A`, `Bx`, `pF`, `pB_F`, `pB_total`, `post_F4` and `m3_results`, because Member 4's validation code uses them and they must not be renamed.

5. Section 5.5: Conditional probability

Question. Are large buildings more likely to need extra AHUs than smaller ones?

Rule. P(B given A) = P(A and B) / P(A) = N(A and B) / N(A), valid because P(A) is greater than 0. It fits because the question compares the extra-AHU rate *inside* two groups, which is exactly what a conditional probability measures. The unconditional P(B) would hide the difference.

The 2 x 2 count table (`counts_AB`) splits all 200 buildings by size and extra AHU:

	B: extra AHU	not B: no extra AHU	Total
A: large (more than 64 zones)	97	1	98
not A: not large	49	53	102
Total	146	54	200

Results

Quantity	Calculation	Value	Plain reading
P(A)	98/200	0.4900	49% of buildings are large
P(B)	146/200	0.7300	73% need extra AHUs
P(A and B)	97/200	0.4850	48.5% are both large and need extra AHUs
P(B given A)	97/98	0.9898	among large buildings, 99% need extra AHUs
P(B given not A)	49/102	0.4804	among not-large buildings, 48% do
Difference	0.9898 - 0.4804	0.5094	size adds about 51 percentage points
P(B given A) / P(B)	0.9898 / 0.7300	1.356	being large makes extra AHUs 1.36 times more likely than average

Two routes give the same answer (and the code asserts it): the definition P(A and B) / P(A), and the direct route of first keeping only the large rows and then averaging `B_extra_ahu` on them.

Threshold check (`threshold_check`). The median 64 is a convention, so I repeated the comparison with other cut-offs:

"Large" means n_zones greater than	Buildings called large	P(B given large)	P(B given not large)
40	171	0.830	0.138
50	139	0.921	0.295
64	98	0.990	0.480
80	69	1.000	0.588
100	45	1.000	0.652

At every cut-off, large buildings are far more likely to need extra AHUs. So the finding does not depend on choosing 64. This answers the likely Q&A question "why the median, and what if another threshold?".

Why it happens (mechanism check). The code proves two facts on the data: (1) `min_ahus_required` equals `ceil(n_zones / 10)` in all 200 rows; (2) B is True exactly when `min_ahus_required` is greater than `ahus_configured`. Configured AHUs range only from 3 to 7, so any building with more than 70 zones needs at least 8 AHUs and always has extra ones: P(B given more than 70 zones) = 1.000 over 89 buildings. The single large building without an extra AHU is seed 100089: 65 zones, 7 AHUs configured, 7 required, 3 floors. It is large, but 7 AHUs already cover 65 zones.

Chart. The bar chart shows three bars: P(B) = 0.730 (grey, everyone), P(B given A) = 0.990 (orange, large), P(B given not A) = 0.480 (green, not large). The unconditional bar sits between the two because it is their weighted average.

Interpretation. Size is a very strong predictor of needing extra AHUs. This is a mechanical rule of the generator (an association in generated data), not a causal discovery about real buildings.

6. Section 5.6: Multiplication rule and chain rule

Question. What share of all buildings is both large and needs extra AHUs, and can it be built from a marginal and a conditional probability? Does the same work for three events (4 floors, large, extra AHU)?

Rules. Multiplication rule: P(A and B) = P(A) x P(B given A) = P(B) x P(A given B). Chain rule: P(F4 and A and B) = P(F4) x P(A given F4) x P(B given F4 and A). It is the right tool because I already have a conditional probability and want the joint one. The plain product

P(A) x P(B) is valid only if the events are independent.

Two events

Route	Calculation	Result
P(A) x P(B given A)	0.4900 x 0.9898	0.4850
P(B) x P(A given B)	0.7300 x 0.6644	0.4850
Direct count	97/200	0.4850
Plain product if independent	0.4900 x 0.7300	0.3577

The first three agree exactly. The plain product (0.3577) does **not** match the true 0.4850, which tells us A and B are **dependent** (Member 4 tests this formally in Section 5.9).

Three events (chain rule)

Factor	Meaning	Calculation	Value
P(F4)	share of 4-floor buildings	63/200	0.3150
P(A given F4)	share of 4-floor buildings that are large	50/63	0.7937
P(B given F4 and A)	share of large 4-floor buildings needing extra AHUs	50/50	1.0000
Product	0.3150 x 0.7937 x 1.0000		0.2500
Direct count	F4 and A and B	50/200	0.2500

Interpretation. 25% of all buildings are 4-floor, large and need extra AHUs. Every large 4-floor building needs extra AHUs. Rule-based and direct values agree exactly because they come from the same 200 rows.

7. Section 5.7: Total probability

Question. Can the overall P(B) be rebuilt from floor-level rates and the floor mix, without counting B over all 200 rows?

Rule. Law of total probability: P(B) = sum over k = 2, 3, 4 of P(B given F_k) x P(F_k). It requires the pieces F2, F3, F4 to be **mutually exclusive** (no building in two groups) and **exhaustive** (every building in some group). The code verifies this: each row has exactly

one floor count, and 74 + 63 + 63 = 200. The rule fits because the extra-AHU rate clearly differs by floor count, so P(B) is a weighted average of group rates. It is also the denominator for Bayes in 5.8.

`total_prob_table`

Floors	N(F_k)	P(F_k)	N(B and F_k)	P(B given F_k)	P(B given F_k) x P(F_k)
2	74	0.370	39	0.5270	0.195
3	63	0.315	49	0.7778	0.245
4	63	0.315	58	0.9206	0.290
Sum	200	1.000	146		**0.730**

Total probability gives 0.195 + 0.245 + 0.290 = **0.7300**, and the direct count gives 146/200 = **0.7300**. They are identical.

Cross-check with a second partition. Using A / not A instead: 0.490 x 0.9898 + 0.510 x 0.4804 = 0.7300. Same result, so the answer does not depend on which valid partition is used.

Joint table (`joint_F_B`): P(floors and B), the pieces of the law, reusable for Member 5's heatmap.

Floors	no extra AHU (not B)	extra AHU (B)
2	0.175	0.195
3	0.070	0.245
4	0.025	0.290

The right-hand column adds up to 0.730 = P(B). The whole table adds up to 1.

Interpretation. The extra-AHU rate rises with the number of floors: 52.7% (2 floors), 77.8% (3 floors), 92.1% (4 floors). More floors means more zones in this generator, so more AHUs are required. The overall 73% is the blend of these three rates according to how many buildings have each floor count. An exact match proves the partition is valid; a mismatch would reveal overlapping or missing groups.

8. Section 5.8: Bayes' theorem

Question. If we only know a building needed extra AHUs, how likely is it to have 4 floors (and to be large)? And what if it needed no extra AHU?

Rule. P(F_k given B) = P(B given F_k) x P(F_k) / P(B), where the denominator P(B) is the total-probability result from 5.7. Bayes fits because the data naturally give "given the floors, how often is there an extra AHU?" but the question asks the reverse. Plain-words vocabulary:

- **Prior** P(F_k): belief about the floor count *before* seeing any evidence.
- **Likelihood** P(B given F_k): how common the evidence is within each floor group.
- **Evidence** P(B): how common the evidence is overall (from 5.7).
- **Posterior** P(F_k given B): updated belief *after* seeing the evidence.

`prior_posterior` table

Floors	Prior P(F_k)	Likelihood P(B given F_k)	Posterior P(F_k given B)	Direct count among B	Posterior P(F_k given not B)
2	0.370	0.5270	0.2671	39/146	0.6481
3	0.315	0.7778	0.3356	49/146	0.2593
4	0.315	0.9206	0.3973	58/146	0.0926

Hand check for 4 floors: (0.9206 x 0.315) / 0.730 = 0.3973, which equals the direct count 58/146. Each posterior column adds up to 1 (the code asserts this), so it is a valid probability distribution.

Second Bayes question. P(A given B) = P(B given A) x P(A) / P(B) = (0.9898 x 0.49) / 0.73 = 0.6644 = 97/146. Among buildings with extra AHUs, about two thirds are large, compared with the prior of 0.49.

Direction warning. P(B given A) = 0.990 and P(A given B) = 0.664 are different questions. The first is high because nearly all large buildings need extra AHUs. The second is lower because many not-large buildings (49 of the 146) also need them. Always say both in words next to every number.

Charts

- *Prior vs posterior bars:* grey = prior, orange = after learning "extra AHU", green = after learning "no extra AHU". Extra AHU moves belief toward 4 floors (0.32 to 0.40) and away from 2 floors (0.37 to 0.27). No extra AHU moves it strongly toward 2 floors (0.65)

and nearly rules out 4 floors (0.09).

- *Probability tree:* the root is all 200 buildings. First branches carry P(F2) = 0.370, P(F3) = 0.315, P(F4) = 0.315. Second branches carry P(B given F_k) on top (0.527, 0.778, 0.921) and P(not B given F_k) below (0.473, 0.222, 0.079). The text at each end is the **joint** probability = product along the path: 0.195, 0.175, 0.245, 0.070, 0.290, 0.025. The six ends add up to 1. Reading the tree left to right is the multiplication rule, adding the "extra AHU" ends is total probability, and dividing one such end by their sum is Bayes.

Interpretation. The posterior is a re-weighting of the observed buildings, not a causal claim: extra AHUs act as evidence about floors only because the generator links floors, size and AHU requirements.

9. Hand-off values for Member 4 (`m3_results`)

All values come live from the cells, nothing is typed by hand.

Key	Meaning	Value
P(B given A)	extra AHU among large	0.9898
P(B given not A)	extra AHU among not large	0.4804
P(A and B)	multiplication rule	0.4850
P(F4 and A and B)	chain rule	0.2500
P(B)	total probability	0.7300
P(F4 given B)	Bayes	0.3973
P(A given B)	Bayes	0.6644
P(F2 given B)	Bayes	0.2671
P(F3 given B)	Bayes	0.3356

Member 4 uses these as the "rule-based" route of his three-way validation (rule vs direct count vs Monte Carlo).

10. Why the numbers agree, and what the assumptions are

- **Exact agreement.** Rule-based and direct values are identical because both are computed from the same finite set of 200 rows. If they ever differed, that would mean a bug, such as a partition that does not add up to 200. Every cell contains `assert` lines that stop the notebook if a check fails.
- **Where differences would appear.** Monte Carlo resampling and the bootstrap (Member 4) differ slightly because of sampling error. That is separate from modelling choices such as treating generator output as random draws.
- **Assumption.** The 200 accepted buildings are treated as equally likely draws from the generator output (not claimed to be a random sample of anything).
- **Limitations.** Results apply only to accepted seeds; n = 200 is small; the 64-zone cut-off is a convention (but 5.5 shows the result survives other cut-offs); the data has no energy or comfort outcome; the relations are generator rules, not real-world causation.

11. Likely Q&A questions and short answers

1. **Why is P(B given A) so different from P(B)?** P(B) averages large and not-large buildings; conditioning removes the not-large group, where the rate is only 0.48. Rule: conditional probability.
2. **Why can't we say size causes extra AHUs?** These are associations in generated data. The link is a generator rule (required AHUs = zones divided by 10, rounded up), not evidence about real buildings.
3. **Why do rule and direct counts match exactly?** Same 200 rows, same population; the formulas are algebraic rearrangements of counting.
4. **Why is the total-probability partition valid?** Every building has exactly one floor count, so F2, F3, F4 are disjoint and cover all 200 buildings.
5. **What is the difference between P(B given A) and P(A given B)?** Different denominators: 98 large buildings versus 146 buildings with extra AHUs.
6. **Prior vs posterior?** The prior is the floor mix before evidence; the posterior is the floor mix among buildings known to need extra AHUs.
7. **Why does the plain product P(A) x P(B) fail?** It assumes independence; the real joint (0.485) is much larger than 0.358, so A and B are dependent.
8. **What if the threshold is not 64?** The threshold table shows the same pattern at 40, 50, 80 and 100 zones.

12. Suggested speaking outline for slides 6 and 7

- **Slide 6 (conditional and multiplication):** the 2 x 2 table; 0.990 vs 0.480 vs 0.730; the product rule giving 0.485 and the independence gap of 0.358.
- **Slide 7 (total probability and Bayes):** the probability tree; 0.195 + 0.245 + 0.290 = 0.730; prior 0.315 to posterior 0.397 for 4 floors; the warning that P(B given A) is not P(A given B).

Before submission, open the notebook in Colab and use Runtime > Restart and run all to confirm it reruns without errors.